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ADVANCEMENT IN THE FIELD OF COMMUNICATION

CDMA and GSM are the names of competing cellular phone standards.

GSM

Global system for mobile communications (GSM), an international standard, ls a digital system using GSM using TDMA (Time Division Multiple Access) technologies. Using TDMA, a narrow band that is 30 kHz wide and 6.7 milliseconds long is split time-wise into three-time slots. Each conversation gets the radio for one-third of the time. This is possible because voice data that has been converted to digital information is compressed so that it takes up significantly less transmission space. Therefore, TDMA has three times the capacity of an analogue system using the same number of channels.

GSM systems provide some useful features:

1. Uses encryption to make phone calls more secure
2. Data networking
3. Short Message Service (SMS) for text messages and paging
4. Call forwarding
5. Caller ID
6. Call waiting
7. Multi-party conferencing

CDMA

Code-division multiple access (CDMA) refers to any of several protocols used in so-called second generation (2G) and third-generation (3G) wireless communications. As the term implies, CDMA is a form of multiplexing, which allows numerous signals to occupy a single transmission channel optimizing the use of available bandwidth. Audio input is first digitized into binary elements. The frequency of the transmitted signal is then made for vary according to a defined pattern (code) so it can be intercepted only by a receiver whose frequency response is programmed with the same code, so it :

follows exactly along with the transmitter frequency. There are trillions of possible frequency sequencing codes, which enhance privacy and makes cloning difficult. CDMA is compatible with other cellular technologies; this allows for nationwide roaming.

Deterrence Between GSM and CDMA

The basic difference between GSM and CDMA is that GSM is specific to a SIM card that is used with the mobile phone. On the other hands, the CDMA is handset specific. Let us discuss some more differences between GSM and CDMA in the comparison chart shown below.

Comparison Chart

BASIS FOR COMPARISON	**GSM**	**CDMA**
Basic	GSM is SIM specific	CDMA is Handset specific
Full Form	Global System for Mobile Communication	Code Division Multiple Access.
Technology	FDMA and TDMA	CDMA
Network	Network tower in each cell serves the mobile phone of that area.	There are a physical channel and a dedicated code for each device in the network.
Transmission	Voice and data transmission at the same time.	Cannot do voice and data transmission simultaneously.
Roaming	Worldwide	Limited
Data Hand	Slower	Faster

GENERAL PACKET RADIO SERVICE (GPRS)

General packet radio service (GPRS) is a packet oriented mobile data service on. the 2G and 3G cellular communication system's global system for mobile communications (GSM), GPRS usage is typically charged based on the volume of data. This contrasts with circuit switching data, which is typically billed per minute of connection time, regardless of whether or not the user transfers data during that period. GPRS is a best-

effort service, implying variable throughput and latency that depend on the number of other users sharing the service concurrently, as opposed to circuit switching, where a certain quality of service (QoS) is guaranteed during the connection.

MOBILE PHONE GENERATIONS

MOBILE RADIO TELEPHONE (OR 0G)

Mobile radio telephone systems preceded modern cellular mobile telephony technology. Since they were the predecessors of the first generation of cellular telephones, these systems are sometimes retroactively referred to as **pre-cellular** (or sometimes zero generation) systems. These mobile telephones were usually mounted in cars or trucks, though briefcase models were also made. Motorola in conjunction with the Bell System operated the first commercial mobile telephone service, Mobile Telephone System (MTS) in the US in 1946, as a service of the wireline telephone company.

1G

IG refers to the first-generation of wireless telephone technology. These are the analogue telecommunications standards that were introduced in the 1980s and continued until being replaced by 2Gdigital telecommunications. The main difference between the two succeeding mobile telephone systems, 1G and 2G, is that the radio signals that 1G networks use are analog, while 2G networks are digital. But both systems use digital signaling to connect the radio towers (which listen to the handsets) to the rest of the telephone system.

2G

2G is short for second-generation wireless telephone technology. 2G cellular telecom networks were commercially launched on the GSM (Global System for Mobile Communication) standard in Finland. Three primary benefits of 2G networks over their predecessors were that phone conversations were digitally encrypted, 2G systems were significantly more efficient on the spectrum allowing for far greater mobile phone penetration levels; and 2G introduced data services for mobile, starting with SMS text messages. 2G technologies can be divided into TDMA (Time Division Multiple Access)-

based and CDMA (Code Division Multiple Access)-based standards depending on the type of multiplexing used.

2.5G(GPRS)

2.5G Is a stepping stone between 2G and 3G cellular wireless technologies, 2.5G (second and a half generation) is used to describe 2G-systems that have implemented a packet-switched domain in addition to the circuit-switched domain. It does not necessarily provide faster services because bundling of timeslots is used for circuit-switched data services (HSCSD) as well. 2.5G provides some of the benefits of 3G (e.g. It is packet-switched) and can use some of the existing 2G infrastructures in GSM and CDMA networks. The commonly known 2.5G technique is GPRS (General Packet Radio Service). GPRS can be used for services such as Wireless Application Protocol (WAP) access, Multimedia Messaging Service (MMS), and Internet communication services such as email and World Wide Web access. GPRS data transfer is typically charged per megabyte of traffic transferred, while data communication via traditional circuit switching is billed per minute of connection time, independent of whether the user is utilizing the capacity or is in an idle state.

2.75G (EDGE)

A 2G mobile phone is a circuit switched digital mobile phone. A 3G mobile is a digital phone with rapid data according to one of the standards being a member of the IMT-2000 family of standards. After those terms were defined, slow packet switched data was added to 2G standards and called 2.5G. 2.75G is the term which has been decided on for systems which don't meet the 3G requirements but are marketed as if they do (e.g. CDMA-200 without multi-carrier) or which do meet the, requirements but aren't strongly marketed as such. (e.g. EDGE systems) Enhanced Data rates for GSM Evolution (EDGE), Enhanced GPRS (EGPRS), or IMT Single Carrier (MT-SC) is a backwards-compatible digital mobile phone technology that allows improved data: transmission rates, as an extension on top of standard GSM.

3G

3G or 3rd generation mobile telecommunications are a generation of standards for mobile phones a mobile telecommunication services fulfilling the International Mobile

Telecommunications 2000 (MT 2000) specifications given by the International Telecommunication Union (ITU). Application service include wide-area wireless voice telephone, mobile Internet access, video calls and mobile TV, all in a mobile environment. 3G was relatively slow to be adopted globally. In some instances, 3G networks do not use the same radio frequencies as 2G, so mobile operators must build entirely new networks and license entirely new frequencies, especially so to achieve high data transmission rates.

The bandwidth and location information available to 3G devices gives rise to applications not previously available to mobile phone users.

Some of the applications are:

1. Mobile TV- a provider redirects a TV channel directly to the subscriber's phone where it can be watched.
2. Video on demand - a provider sends a movie to the subscriber's phone.
3. Videoconferencing —subscribers can see as well as talk to each other.
4. Telemedicine - a medical provider monitors or provides advice to the potentially isolated subscriber.
5. Location-based services - a provider sends localized weather or traffic conditions to the phone or the phone allows the subscriber to find nearby businesses or friends.

3.5G

High-Speed Downlink Packet Access or HSDPA is a mobile telephony protocol. Also called 356. HSDPA is a packet-based data service in W-CDMA downlink with data transmission up to 8-10 Mbit/s (and 20 Mbit/s for MIMO systems) over a 5MHz bandwidth in WCDMA downlink. HSDPA I implementations include Adaptive Modulation and Coding (AMC), Multiple-input Multiple-Output (MMO), Hybrid Automatic Request (HARQ), fast cell search, and advanced receiver design.

4G

4G is the fourth generation of cell phone mobile communications standards. A 4G system provides mobile ultra-broadband Internal access, for example to laptops with

USB wireless moderns, to smartphones, and other mobile devices. Conceivable applications include amended mobile web access, IP telephony, gaming services, high-definition mobile TV. Video conferencing and 3D television.

In March 2008, the International Telecommunications Union-Radio communications sector (ITU-R) specified a set of requirements for 4G standards, named the International Mobile Telecommunications Advanced (IMT-Advanced) specification, setting peak speed requirements for 4G service at 100 megabits per second (Mbit/s) for high mobility communication (such as from trains and cars) and 1 gigabit per second (Gbit/s) for low mobility communication (such as pedestrians and stationary users).

As opposed to earlier generations, a 4G system does not support traditional circuit-switched telephone service, but all-internet Protocol (IP) based communication such as IP telephony. Unlike 3G, which is based on two parallel infrastructures consisting of circuit switched and packet switched network nodes respectively, 4G will be based on packet switching only. This will require low-latency data transmission.

Long Term Evolution (LTE) is the latest rolled-out wireless mobile broadband technology that is powering 4G, networks designed primarily for data transmission at unprecedented speeds. It uses the spectrum to carry data traffic. Since LTE uses wider chunks of spectrum, data speeds on LTE based 4G networks are nearly four times faster than on 3G. LTE is also a scalable bandwidth technology that works alongside 2G and 3G, So, a 3G operator can easily upgrade his network to LTE.

5G

5G is the fifth generation of cell phone mobile communications standards after 4G LTE networks. In early 2012, ITU Initiated the development of "IMT for 2020 and beyond", setting the stage for 5G research activities and establishing the requirements and vision for 5G. Under ITU's IMT-2020 programme, ITU membership is developing the international standards to achieve well-performing 5G networks.

The first full-scale commercial deployments for 5G are expected shortly after IMT-2020 specifications are finalized in 2020. ITU has reached first-stage approval (consent) of three new-international standards defining the requirements of IMT-2020

(5G) systems as they relate to network operation, softwarization and fixed-mobile convergence.

Like previous mobile broadband generation, 5G networks will use the radio-frequency spectrum. The radio frequency spectrum is divided into frequency bands, allocated to radiocommunication services in such a way that each band may be used only by services which can co-exist with each other. The increased traffic and speed required for 5G will necessitate much more spectrally efficient technologies, as well as a lot of additional spectrum, beyond what is currently used by 3G and 4G.

Most of it will come from frequency bands above 24 GHz, which pose considerable challenges regarding radio wave propagation and are also used by some radiocommunication services, notably for satellite communications, weather forecasting and monitoring of Earth resources and climate change.

The data rate has improved from 64 kbps in 2G to 2Mbps in 3G and 50-100 Mbps in 4G, 5G is expected to enhance not only the data transfer speed of mobile networks but also the scalability, connectively, and energy efficiency of the network. It is assumed that by 2020, 50 billion devices will be commercial to the global IP network, which would appear to present a challenge. In a true 'networked society" remote controlled operation of appliances and critical commercial machines over a reliable 5G network will be possible with zero delays. Real-time control of machines by using mobile devices will be possible, making the internet of things (IoT) more available to all.

Finally, but not least important, less energy-hungry network nodes will be required towards a greener world. Therefore, the following are the most important elements in the description of 5G. High throughout, low-latency, high reliability, increased scalability, and energy efficient mobile communication technology.

5G is expected to support applications such as smart homes and buildings, smart cities, 3D video, work and play in the cloud, remote medical surgery, virtual and augmented reality, and massive to machine communications for industry automation and self-driving cars. 3G and 4G networks currently face challenge in supporting these services.

Questions:

1. Write a short note on CDMA & GSM and its difference.
2. Enumerate the evolution of mobile phone generations stage by stage.
3. Write a short note on 5G.